



Editorial

Bilateral uveal melanoma: hunting for the source

Bilateral uveal melanoma is extremely rare.^{1–9} In an analysis of 8033 consecutive patients with uveal melanoma managed over 38 years from a tertiary referral centre, there were 11 (<1%) bilateral cases.¹ The bilateral cases occurred in age groups 21–60 years (6/4281, <1%) and >60 years (5/3635, <1%). Melanoma-related metastasis in the entire cohort was found occurring in 15% at 5 years, 25% at 10 years and 36% at 20 years.¹

Specific information regarding bilateral uveal melanoma has been published in single case reports or small case series.^{2–9} Historically, the first bilateral uveal melanoma case published in the United States was reported in 1959 by Wiesinger *et al.*² It was not until 1977 that the second case was published in the United States by Shammass and Watzke, and at that time, there were four published European cases.³ They described a 55-year-old female who developed sequential choroidal melanoma over 4 years and was eventually managed with enucleation of both eyes. They claimed that bilateral choroidal melanoma was estimated to occur in the United States once every 18 years at a rate of 1 in every 50 million Caucasians.³

What causes unilateral uveal melanoma? It is speculated that some unilateral uveal melanomas arise from preexisting benign nevus, some from preexisting congenital ocular melanocytosis and others occur *de novo* without a precursor lesion. Regarding choroidal nevus, this common benign tumour is estimated to show a prevalence of at least 5% in the US population, most commonly in Caucasians.¹⁰ If we presume that all uveal melanomas arise from preexisting nevus, then transformation into melanoma is extremely low at a mathematically estimated rate of 1 in 8845, peaking at age 70–74 years, with 1 in 3238 demonstrating growth to melanoma.¹¹ Conditions like ocular melanocytosis can also demonstrate risk for development of uveal melanoma, with a lifetime estimate at 1 in 400.¹² Bilateral ocular melanocytosis has been observed to cause bilateral uveal melanoma,⁴ and a review of

20 published cases of multifocal uveal melanoma in a single eye revealed three (15%) with congenital ocular melanocytosis.¹³ These findings imply that melanocytosis could be an important precursor to unilateral, bilateral and multifocal melanoma, and the authors¹³ implied that other reports may have overlooked subtle pigmentation of this precursor lesion.

What causes bilateral uveal melanoma? This answer is less clear, but there is strong suspicion that some cases could be related to a germline mutation. Several years ago, our team studied eight cases of bilateral uveal melanoma in a cohort of 4500 patients,⁸ and we admitted that this frequency was significantly greater ($P < 0.0001$) than predicted by the population estimates of Shammass and Watzke.³ We observed that the presentation of bilateral disease was typically sequential at a median interval of 10 years between the two eyes, and we noted that melanocytosis was significantly more common in bilateral compared to unilateral melanoma ($P = 0.001$), but this finding did not explain every case. There was no relationship to underlying syndromes such as neurofibromatosis type 1, Li-Fraumeni syndrome or familial atypical mole and melanoma syndrome. In that report, our final concluding statement read: 'Unidentified germline mutations may be involved in [the] pathogenesis of bilateral primary uveal melanoma'.⁸

As predicted in our final sentence, new genetic information has come to light and has implicated another source for bilateral uveal melanoma, namely, BAP1 (BRCA - associated protein 1) tumour predisposition syndrome.^{14,15} This germline cancer syndrome, identified recently, can lead to several cancers, including uveal melanoma, and follows an autosomal-dominant inheritance pattern. The spectrum of tumour types, even within a single family, can vary.¹⁴ According to Rai *et al.*, analysis of 174 cases of patients with germline BAP1 mutation revealed the most common tumours to be uveal melanoma (31%), mesothelioma (22%), cutaneous

Conflict of interest: None declared.

Funding sources: This study was supported by the Eye Tumor Research Foundation, Philadelphia, PA (CLS). The funders had no role in the design and conduct of the study; in the collection, analysis and interpretation of the data; or in the preparation, review or approval of the manuscript. CLS, MD, has had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

melanoma (13%), renal cell carcinoma (10%), breast carcinoma (10%), atypical Spitz tumour (18%) and others.¹⁴ Subsequent publications have found BAP1 germline mutation in patients with familial uveal melanoma,¹⁶ multifocal uveal melanoma in a single eye¹⁷ and now bilateral uveal melanoma.¹⁸

In this report by Scott *et al.*, a population-based study and systematic review was performed using three sources: a pathology database from 1996 to 2016; the Surveillance, Epidemiology and End-Results (SEER)-18 from 1973 to 2103; and published English literature, resulting in a total of 52 cases of bilateral uveal melanoma.¹⁹ Genetic testing was not a part of this analysis. They noted that 69% of bilateral cases presented with unilateral tumour and later developed melanoma in the second eye at a median interval of 22 months. In addition, the first affected eye was more likely to be managed with enucleation ($P = 0.002$), whereas the second eye was salvaged with conservative therapy. They found that survival did not differ between patients with unilateral *versus* bilateral uveal melanoma; however, they acknowledged that the number of bilateral cases were few. These authors astutely speculated that the observation of 'uveal melanoma occurring in a paired organ suggests a genetic predisposition' to developing this malignancy.¹⁹

Based on recent data regarding BAP1 tumour predisposition syndrome in uveal melanoma patients with various heritable scenarios, we speculate that some patients with bilateral uveal melanoma could demonstrate germline BAP1 mutation. Furthermore, such cases with BAP1 mutation could display an increased risk for metastasis. In a comparative study of 507 patients with uveal melanoma who were tested for BAP1 germline mutation, Gupta *et al.* found that metastatic risk was significantly greater with BAP1 germline mutation (71%) *versus* those without mutation (18%) ($P = 0.003$).²⁰

In summary, we suggest that all patients with bilateral uveal melanoma and those with multifocal or familial uveal melanoma; those with melanoma at a young age; and those with personal/family history of mesothelioma, cutaneous melanoma, renal cell carcinoma and other cancers should be tested for germline BAP1 tumour predisposition syndrome. Patients identified as positive should have relatives tested and should maintain routine screening for related cancers.

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